Halstead Metrics Analysis Matrix

Subject: Digital Document Verification System (DDVS) | **Artifact measured:** src/ddvs_core.py (reference implementation of registration, hashing and verification logic, SRS FR-13 to FR-36, FR-48) | **Size:** 84 lines, 69 non-blank/non-comment lines

1. Measurement method

Metrics are computed by tools/halstead.py (reproducible: `python tools/halstead.py src/ddvs_core.py`). **Operators** = Python keywords (except True/False/None) and operator/punctuation tokens, each bracket pair counted once via its opening token. **Operands** = identifiers, numeric and string literals, True/False/None. Comments and docstrings are excluded. Function rows count tokens inside that function's line range (including its signature); the module row additionally includes imports and module-level constants, so it is larger than the sum of the function rows.

Formulas: n = n1 + n2; N = N1 + N2; estimated length N^ = n1 log2 n1 + n2 log2 n2; Volume V = N log2 n; Difficulty D = (n1 / 2) x (N2 / n2); Effort E = D x V; Time T = E / 18 seconds; Delivered bugs B = V / 3000. n1/n2 = distinct operators/operands; N1/N2 = total operators/operands. CC = McCabe cyclomatic complexity (radon).

2. Metrics matrix

Function / Unit	n1	n2	N1	N2	n	N	N^	Volume V	Difficulty D	Effort E	Time T (s)	Bugs B	CC
compute_hash	5	5	8	6	10	14	23.22	46.51	3.0	139.52	7.75	0.0155	1
generate_verification_id	7	7	12	7	14	19	39.3	72.34	3.5	253.19	14.07	0.0241	1
validate_upload	14	17	29	24	31	53	122.79	262.57	9.88	2594.83	144.16	0.0875	4
register_document	15	29	65	56	44	121	199.48	660.59	14.48	9567.18	531.51	0.2202	6
verify_document	18	32	57	51	50	108	235.06	609.54	14.34	8743.04	485.72	0.2032	7
verify_by_file	13	24	42	37	37	79	158.14	411.55	10.02	4124.04	229.11	0.1372	3
revoke_document	9	7	12	11	16	23	48.18	92.0	7.07	650.57	36.14	0.0307	2
MODULE (ddvs_core.py)	27	87	238	207	114	445	688.92	3040.64	32.12	97667.33	5425.96	1.0135	-

3. Interpretation

- **Highest effort:** register_document (E = 9,567), then verify_document (E = 8,743) and verify_by_file (E = 4,124). These three hold the validation and decision logic, so they are the first candidates for focused testing and review.

- **Highest difficulty:** register_document (D = 14.48; n1 = 15 distinct operators, average operand reuse N2/n2 = 1.93).

- **Module totals:** Volume = 3,041, Effort = 97,667, estimated delivered bugs B = 1.0135 (about one latent defect per 3000 units of volume - a coarse estimate, not a guarantee).

- **Complexity:** maximum cyclomatic complexity is 7 (verify_document: format check, four result branches and two boolean conditions), well inside the commonly used threshold of 10, so no function needs to be split. Radon Maintainability Index = 40.15 (rank A, scale 0-100).

- **Test link:** all 12 unit tests in tests/test_ddvs_core.py pass (python -m unittest tests.test_ddvs_core) and cover every verification outcome (Verified, Invalid, Not Found, Tampered, Malformed ID), registration validation, ID uniqueness and logging of every attempt (FR-37).

- **Scope note:** Halstead is defined over source code, so these figures measure the reference implementation of the core module only, not the full web application (UI, REST layer and database access are outside this file).